P116: Order Book Dynamics Around Major Events

Progress and discussion agenda

1. **Progress to date**
   - Explore the platform and establish the data workflow
   - Select assets and macroeconomic announcements
   - Collect daily futures data and define contract selection
   - Compare daily moves on announcement and normal days
2. **Next research steps**
   - Analyse intraday dynamics using market-by-order (MBO) data
   - Discuss a dashboard for monitoring market conditions

Asset universe

Futures offer liquid benchmark exposures, long trading hours and a common exchange data source.

ZQ: Fed funds	Current calendar month and next six months; expected average overnight rates and policy-path repricing.
ZT: 2-year Treasury	Short-rate expectations and demand for safe assets.
ZN: 10-year Treasury	Longer-run rate expectations and term premium; complements ZT in studying the curve.
GC: Gold	Responses to real rates, the dollar and demand for safe assets.
6E: EUR/USD	Dollar-relative currency response; higher prices mean a stronger euro.
ES: S&P 500	Equity responses to rates, earnings expectations and growth news.
BTC: Bitcoin	Compare a newer asset's response with established markets.

All contracts are available through Databento's `GLBX.MDP3` dataset.

Daily data and contract selection

- Retrieve explicit futures contracts using Databento's `ohlcv-1d` schema.
- **Fed funds:** retain seven calendar-month horizons, T0–T6, on each date.
- **Other assets:** retain two nearby contracts and mark one as primary.
- Switch primary when the challenger accounts for **at least 65% of their combined volume for three consecutive trading observations**.
- If only one contract has a bar, use that contract immediately; if neither has a bar, neither is primary.
- Compute each open-to-close return within one contract. No return is calculated across a roll-price gap.

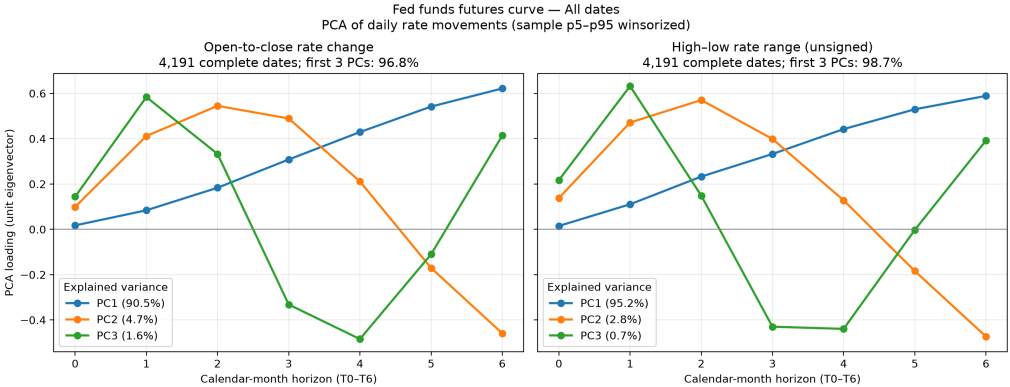
Daily analysis before the intraday study

Coverage: June 2010–October 2026; Bitcoin has a shorter history. Events: FOMC, CPI, nonfarm payrolls (NFP), and PCE releases.
Time window: bars span midnight to midnight UTC, crossing CME session boundaries. Intraday research will align observations to release timestamps.

1. **Same-day correlations:** compare signed open-to-close moves across assets and event samples.
2. **Lag-one autocorrelation:** correlate today's move with the next observed trading day's move; condition on the first day being an event.
3. **Range intensity:** divide today's high–low range by its preceding 60-trading-date mean, excluding today.

FF uses PCA of seven rate changes or ranges. Other signed returns are $(C - O)/O$. Normal days exclude all four event categories.

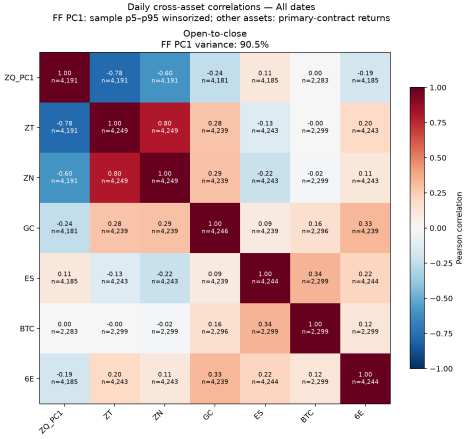
Fed funds curve PCA



Covariance PCA: demean each maturity after p5–p95 winsorization; no scaling by maturity standard deviation. Eigenvectors have unit length.

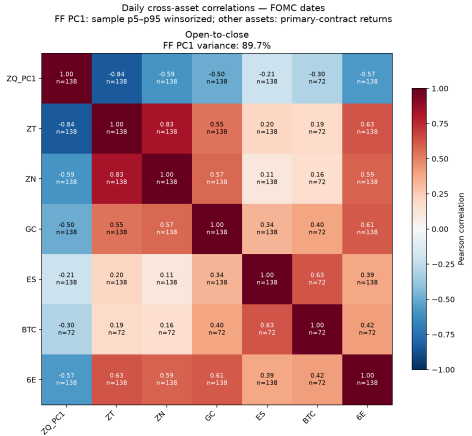
Observation: PC1 is a ramp rather than a parallel shift. Further horizons carry more movement variance; current-month averaging may also suppress the front end.

Cross-asset correlations: all dates



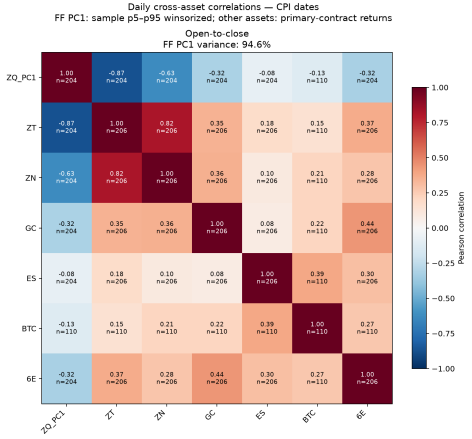
Observation: The rates group is strongly linked: ZT–ZN = 0.80; FF rate-change PC1–ZT = −0.78. All dates include event days.

Cross-asset correlations: FOMC dates



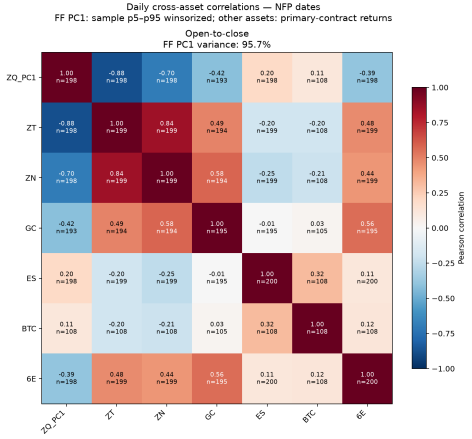
Observation: Several relationships strengthen: BTC–ES 0.34 to 0.63; gold–EUR 0.33 to 0.61; BTC–EUR 0.12 to 0.42. BTC pairs have 72 FOMC observations.

Cross-asset correlations: CPI dates



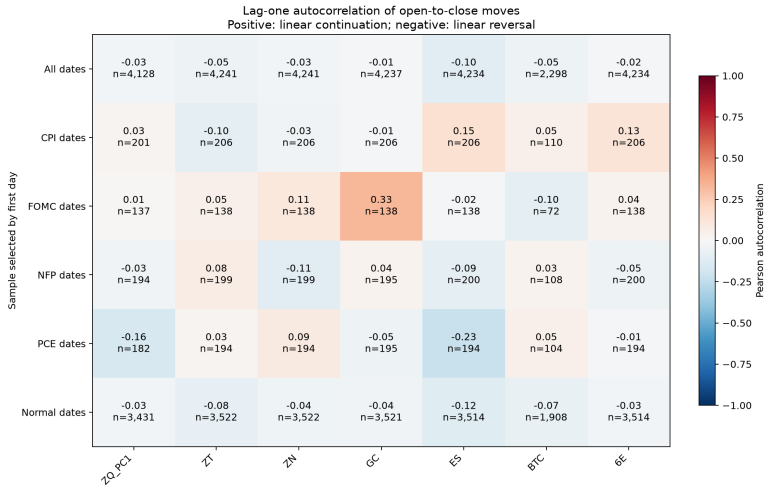
Observation: FF PC1–ZT $= -0.87$; gold–EUR $= 0.44$ and BTC–ES $= 0.39$. The FOMC pattern is not shared uniformly across releases.

Cross-asset correlations: NFP dates



Observation: Strong links span rates, gold and FX: ZN–gold = 0.58 and gold–EUR = 0.56. BTC–ES = 0.32, close to its full-sample value.

Lag-one autocorrelation of daily moves



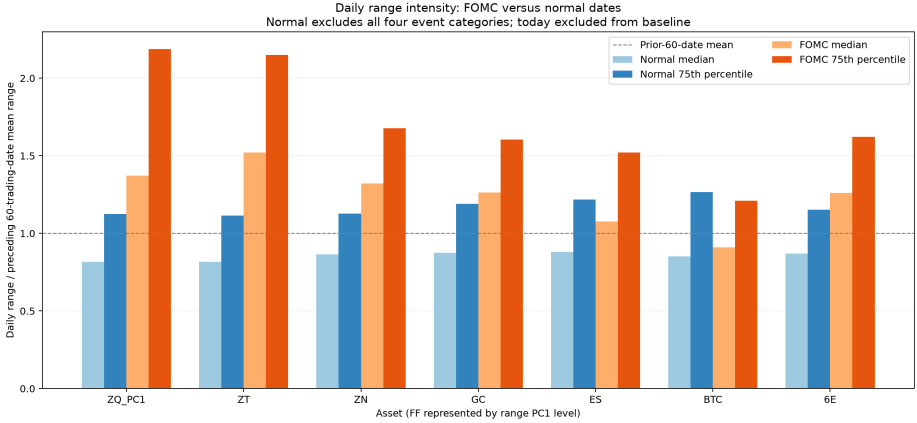
Compare: day t open-to-close with day $t+1$ open-to-close. Select samples by day t; day $t+1$ may be any event category.

Gold stands out: FOMC $\rho = 0.33$, versus -0.04 on normal days.

Positive values indicate linear continuation relative to sample means, not a guaranteed repeated sign.

FF uses a common full-sample PC1 basis directly, without differencing PC1 again.

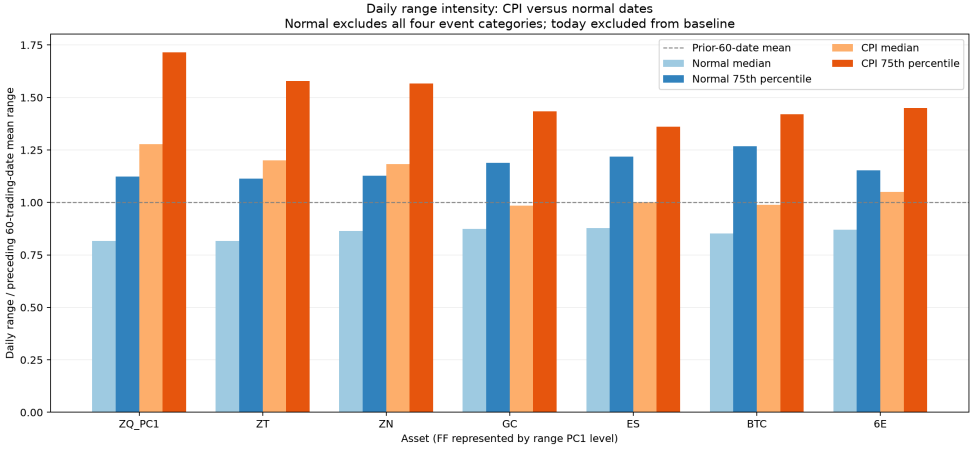
Daily range intensity: FOMC dates



$I_t = R_t / \text{mean}(R_{t-60}, \ldots, R_{t-1})$. For FF, R_t is the uncentered projection of clipped seven-horizon ranges onto range-PC1 loadings.

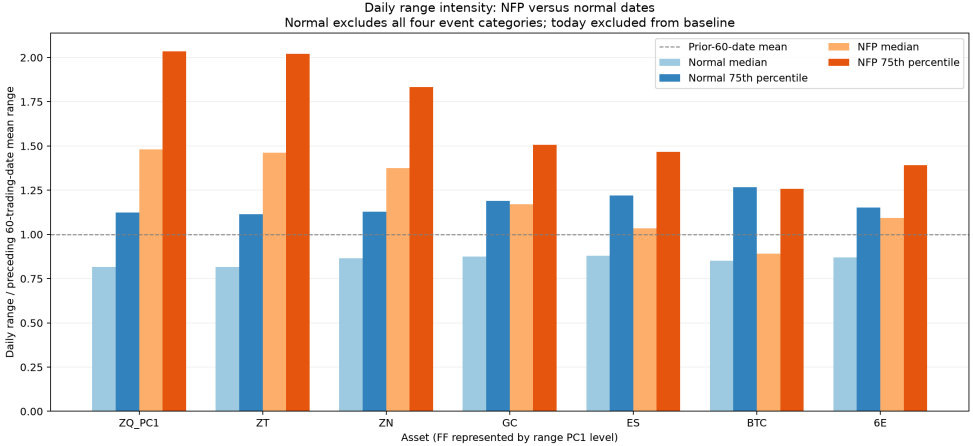
Observation: The largest median increases are in FF and ZT. BTC shows little range expansion despite stronger equity co-movement. This is a range proxy, not a standard-deviation estimate.

Daily range intensity: CPI dates



Observation: Rate futures show elevated median ranges. Gold, ES and BTC medians are near their trailing means, but their upper quartiles exceed normal-day levels.

Daily range intensity: NFP dates



Compare the median and upper quartile with normal days, using the same preceding 60-date baseline. A value of 1 means today's range equals its trailing mean.

Proposed intraday research

Liquidity

- Bid–ask spread
- Depth at level k
- Order additions and cancellations by level

Order flow

- Total traded volume
- Signed trade flow
- Arrival and cancellation rates

Prices

- Returns around release time
- Exponentially weighted volatility
- Price impact of order flow

These are proposals for discussion; prioritisation will depend on data coverage and reconstruction cost.

Study design and key metrics

1. **Event-aligned trajectories**
   - Compare pre- and post-release features with normal days at matching times.
   - Measure time to peak or trough and time to stabilisation.
   - Compare recovery speeds across assets and announcement types.
2. **Information flow across markets**
   - Study lead–lag relationships in prices, order flow and liquidity.
   - Does EUR futures order flow precede gold returns or gold order flow?
   - Does liquidity withdrawal in one market precede withdrawal elsewhere?
   - Explore event-specific flow hierarchies: which markets lead, which follow, and whether that ordering differs across announcements.

Check clock alignment, influential days and stability across periods before interpreting apparent leads as predictive relationships.

Dashboard proposal

- Monitor spreads, depth, signed order flow and price volatility across the chosen markets.
- Compare current conditions with historical event and normal-day patterns.
- Condition reference trajectories on the release type and elapsed time since the announcement.
- Explore expected time to normalisation, with updates as observations arrive.

Scope for discussion: prioritise the empirical intraday study; build a dashboard if time and live-data access permit.
Feedback welcome: which assets, features and event comparisons should come first?

Thank you